

Total No. of Questions : 8]

SEAT No. :

PC-1759

[Total No. of Pages : 2

[6353]-77

T.E. (Electronics / E & TC)
DATABASE MANAGEMENT
(2019 Pattern) (Semester - I) (304183)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Make Suitable assumptions wherever necessary.
- 5) Use of Calculator is allowed.

- Q1)** a) Explain with example : Aggregate functions (any 3), String functions (any 3) **[6]**
- b) Explain any two transaction control commands. **[6]**
- c) What are the different types of joins in SQL? Explain one join with suitable example. **[6]**

OR

- Q2)** a) With the help of block diagram, describe PL / SQL block structure. **[6]**
- b) Explain the following operations with suitable queries. **[6]**
- i) Set operations (any two) ii) Date functions (any two)
- c) Explain what do you mean by DDL, DML and DCL in SQL. **[6]**

- Q3)** a) What is database transaction management? Write transaction states. **[6]**
- b) Explain how deadlock occurs? Which are the actions required for the deadlock recovery process? **[6]**
- c) What are ACID properties of a transaction? **[5]**

OR

- Q4)** a) Explain need and role of Time-Stamp based protocols in database management control. **[6]**

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- b) Explain serializability with respect to conflict and view. [6]
- c) Define the following terms. [5]
 - i) Concurrency
 - ii) Timestamp
 - iii) Timestamp ordering
 - iv) Schedule
 - v) Transaction.

- Q5)** a) Explain the terms speed up and scale up in parallel database. [6]
- b) What are different parallel database architectures? Explain its advantages. [6]
- c) Draw and explain memory structure of instance in oracle architecture. [5]

OR

- Q6)** a) Write short note on centralized database system. [6]
- b) Explain the intra query parallelism query evaluation technique with one example. [6]
- c) Explain Virtualization in multi core processor. [5]

- Q7)** a) Draw and explain client-server architecture for DDBMS. [6]
- b) Compare homogeneous and heterogeneous distributed database. [6]
- c) Explain data replication in distributed data storage. [6]

OR

- Q8)** a) Explain the distributed database system failure modes (any two). [6]
- b) Discuss in detail about single -lock-manager approach in concurrency control. [6]
- c) Write the types of data fragmentation and explain horizontal fragmentation with one example. [6]

